Are the neurons of a reward-trained Ms. Pac-Man agent already clean? A preregistered neuron audit with causal steering

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Abstract

Goodfire's essay "Neural Networks Are Already Clean" argues that most individual neurons of trained networks have one main job. We test this claim, and whether single units can steer behaviour, in three deep RL agents that learned Atari Ms. Pac-Man from game reward alone. They are public Sample Factory APPO agents with a Nature-CNN torso, trained for 2B environment steps.

We re-implement the agents in PyTorch, bit-exact against the original stack (max logit difference 1.1e-05). We decode exact game state from the console RAM, including two frightened-mode addresses that we identify (97.1% agreement with sprite colours), and label every unit against 40 ground-truth features under a preregistered protocol.

Not "already clean". On held-out games, the fraction of alive fc neurons with one main job is 0.8%, vs 0.2% for random directions and 9.4% for an untrained network. The preregistered privileged-basis test is supported for agent 1111, but the effect is trivially small. It is not supported for agent 2222 and not supported for agent 3333. Neurons are instead conjunctive. Exploratorily, Ms. Pac-Man's position × heading explains a median R^2 of 0.20 per neuron, vs 0.03 for the best single feature.

One unit, a no-go zone. Two steering candidates were frozen before any held-out data. Setting a single conv3 unit (channel 18 at one spatial position) to its 99th percentile makes the agent avoid the corresponding maze region: approach rate 0.424 → 0.361, Holm p 0.0002. This effect is stronger than that of each of the 63 other channels at that position (the preregistered H2 passes). Ablating fc neuron 453, which votes UP, cuts moving up toward edible ghosts but does not significantly reduce ghosts eaten per power pill (H2 fails for this candidate). Its "edible ghost is up" label does not survive confound control.

Replication. Frozen selection rules applied to the other two agents give the following (H3):

- no-go-zone rule: replicates;
- ghost-hunting ablation rule: does not replicate.

All tests, deviations and negative results are reported.

1. Introduction

Motivation. Mechanistic interpretability asks whether the internal units of a network can be read as human-understandable variables and whether manipulating them changes behaviour in the predicted way.

The claim we test. Goodfire recently argued that most neuron poles of large vision and language models "have one main job", based on top-activating examples, feature visualisation and steering (Cammarata/Goodfire 2026).

Why RL is a sharp test. Reinforcement-learning agents trained from reward alone offer two advantages:

- the environment state is fully known (for Atari, the console RAM);
- behaviour can be measured causally in closed loop.

Gap in prior work. Prior interpretability work on Ms. Pac-Man used:

- input saliency (Greydanus et al. 2018);
- attention maps (Mott et al. 2019);
- NMF concepts (Bekkemoen & Langseth 2025);
- a sparse autoencoder without interventions (DuPlessie 2024).

Single-channel steering of RL agents has been shown for a Procgen maze agent (Mini, Turner et al. 2023) and for Sokoban planners (Taufeeque et al. 2024; Bush et al. 2025). To our knowledge, no published work has steered individual neurons of a reward-trained Ms. Pac-Man agent.

Contributions.

1. A bit-exact, dependency-light re-implementation of the public Sample Factory Ms. Pac-Man agents with a golden fidelity test.
2. RAM ground truth for Ms. Pac-Man, including the frightened-mode timer `ram[116]` and per-ghost edible bits (bit 7 of `ram[1+i]`), and a maze lattice graph.
3. A preregistered Goodfire-style audit of every fc neuron against three controls (H1).
4. Preregistered causal steering tests with matched random-unit nulls on held-out games (H2), with replication rules for two more agents (H3).

2. Background and related work

- **Neuron-level interpretability.**
  - Circuits (Olah et al. 2020).
  - Polysemanticity and superposition, which motivate dictionary learning (Bricken et al. 2023).
  - Goodfire's counter-argument that neurons are mostly clean (2026).
- **RL interpretability.**
  - An Atari model zoo with activation analysis (Such et al. 2019).
  - CoinRun attribution and model editing (Hilton et al. 2020).
  - Probe-based steering of planning in Sokoban (Bush et al. 2025).
- **Ground truth for Atari.**
  - AtariARI RAM annotations (Anand et al. 2019).
  - OCAtari object extraction (Delfosse et al. 2023).
- **Evaluation protocol.** Sticky actions (Machado et al. 2018).

3. Method

3.1 Agents and fidelity

- **Agents.** `edbeeching/atari_2B_atari_mspacman_{1111,2222,3333}`:
  - Sample Factory 2.0 APPO, 2B env steps, clipped reward, $\gamma = 0.99$, no RNN;
  - Nature-CNN torso (conv 32/64/64) → fc 512 ReLU → linear policy (9 actions) and value heads.
- **Environment.** The wrapper chain is `MsPacmanNoFrameskip-v4` → no-op reset → max-and-skip 4 → episodic life → reward clipping → 84×84 grayscale → stack 4.
- **Fidelity.** We reproduce the wrapper chain and network in PyTorch and verify both against a golden trajectory generated with sample-factory 2.0.3, gym 0.26.2 and ale-py 0.8.1:
  - 0 mismatched observations in 446 steps;
  - max logit difference 1.1e-05.

  Appendix A has the details.

3.2 Ground truth

- **RAM decoding** follows OCAtari and AtariARI.
- **Frightened mode** was identified by correlating RAM bytes with blue ghost pixels and power-pill events.
- **Maze.** A 14×18 lattice graph built from the frame gives breadth-first-search distances and directions.
- **Features.** 40 features in five families (Appendix B). Steps where Ms. Pac-Man does not move (death animation, ready screen, level transition) are excluded.

3.3 Data and stochasticity

- **Splits.** 100 discovery games (seeds 0–99) and 100 held-out games (seeds 10000–10099) per agent.
- **Action sampling.** Actions are sampled from the policy with a per-game RNG, so steered and baseline games share random numbers.
- **Stochasticity fallback.** A preregistered check found that without sticky actions, same-start games stayed identical for a median of 1230 steps. The preregistered fallback (sticky actions, p = 0.25) was therefore used for all data (median divergence 320 steps).

3.4 Audit (H1)

- **Per-unit score.** For each alive fc unit (positive pole = post-ReLU activation):
  - R^2_1, the out-of-sample R^2 of 20-bin means of each single feature;
  - R^2_all, the out-of-sample R^2 of gradient boosting on all features.
- **One-job criterion.** R^2_all ≥ 0.3 and max R^2_1 ≥ 0.7 · R^2_all.
- **Controls.**
  - (a) random orthonormal directions;
  - (b) non-negative mixtures of 2–8 units;
  - (c) an untrained, orthogonally initialised network on the same states.
- **Test.** Game-cluster bootstrap of the difference in one-job fraction. H1 is supported if the 95% CI against (a) excludes 0.

3.5 Steering (H2) and replication (H3)

- **Why intervene upstream of fc.** Because the heads are linear and there is no recurrence:
  - clamping an fc unit only adds a constant action bias;
  - ablating it removes a state-dependent vote;
  - conv3 interventions act through 512 ReLUs.
- **Candidate selection** (discovery data only):
  1. Exact open-loop screens, using a rank-1 update for single conv3 units.
  2. Closed-loop pilots on discovery seeds.
  3. Two candidates frozen in a committed addendum.
- **Confirmatory design** (held-out seeds):
  - 100 baseline games, 50 games per candidate, and 20 games per null unit;
  - nulls: 99 random alive fc units, or the 63 other conv3 channels at the same cell;
  - pass criteria: predicted sign with Holm-corrected paired sign-flip p < 0.05, **and** an effect beyond the 95th percentile of the null.
- **Replication (H3).** Deterministic selection rules transfer the procedure to agents 2222 and 3333.

4. Results

4.1 Fidelity and ground truth

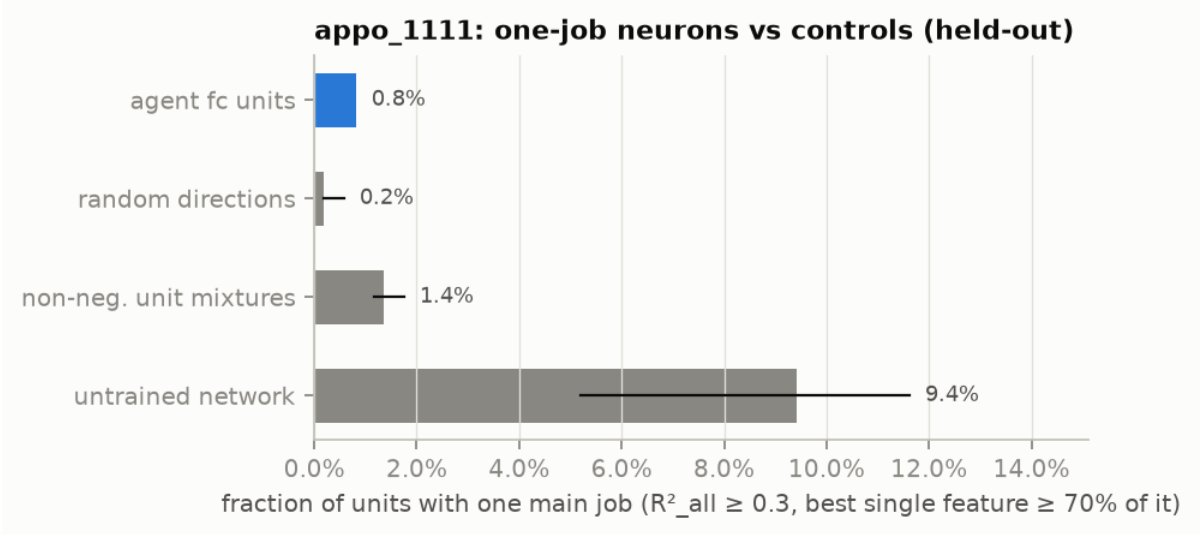
check	value
golden trajectory steps (incl. 2 life losses and game over)	446
steps with any observation pixel mismatch	0
max abs. logit difference	1.1e-05
max abs. value difference	4.8e-07
reference stack	sample-factory 2.0.3, gym 0.26.2, ale-py 0.8.1, torch 1.13.1 (WSL)

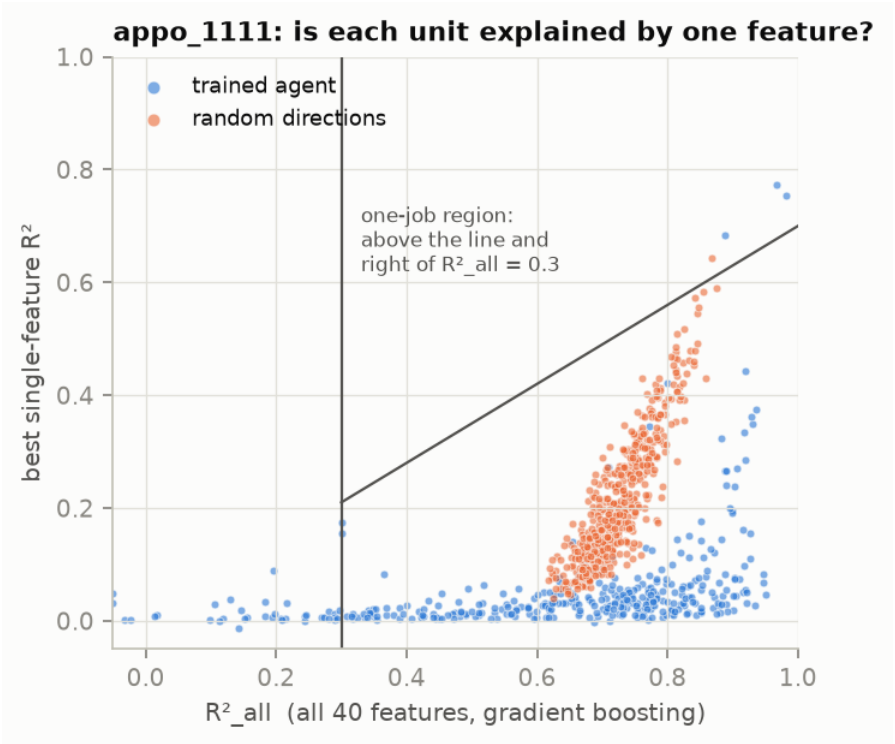
validation (discovery games, sticky actions)	value
agent steps checked	161995 (100 games)
RAM edible bit (bit 7 of ram[1+i]) vs blue sprite pixels, per ghost-step	97.1%
frightened timer ram[116] > 0 vs any ghost edible	100.0%
eaten pellets at Ms. Pac-Man's lattice node (26825 events)	100.0%
steps in control (moving; excludes death, ready and level transitions)	85.1%
distinct maze layouts seen	4

4.2 H1: one-job neurons

unit set (held-out, confirmatory)	units	one-job fraction [95% CI]	median R^2_all	median best single R^2
trained agent (fc units)	367	0.8% [0.8%, 0.8%]	0.708	0.031
random directions (control a)	512	0.2% [0.2%, 0.6%]	0.719	0.185
non-negative unit mixtures (control b)	512	1.4% [1.2%, 1.8%]	0.720	0.053
untrained network (control c)	499	9.4% [5.2%, 11.6%]	0.537	0.096

H1 test (agent minus random directions): 0.6% [0.2%, 0.6%]; H1 supported.





Robustness. The CIs below come from a start-state-cluster bootstrap. The threshold sweep shows that agent 1111's small advantage over random directions reverses at a looser ratio threshold: at a ratio of 0.5, random directions score 8.0% vs the agent's 1.4%.

unit set	one-job fraction	game-bootstrap 95% CI	start-state-cluster 95% CI
trained agent (fc units)	0.8%	[0.8%, 0.8%]	[0.8%, 0.8%]
random directions (control a)	0.2%	[0.2%, 0.6%]	[0.2%, 0.4%]
non-negative unit mixtures (control b)	1.4%	[1.2%, 1.8%]	[1.4%, 2.0%]
untrained network (control c)	9.4%	[5.2%, 11.6%]	[4.8%, 11.0%]

Threshold sweep, one-job fraction shown as agent / random directions (rows: R^2_all threshold; columns: best-single / R^2_all ratio threshold). Start-state clusters: 4.

R^2_all ≥	ratio ≥ 0.5	0.6	0.7	0.8	0.9
0.1	1.6% / 8.0%	0.8% / 1.6%	0.8% / 0.2%	0.0% / 0.0%	0.0% / 0.0%
0.2	1.6% / 8.0%	0.8% / 1.6%	0.8% / 0.2%	0.0% / 0.0%	0.0% / 0.0%
0.3	1.4% / 8.0%	0.8% / 1.6%	0.8% / 0.2%	0.0% / 0.0%	0.0% / 0.0%
0.4	1.1% / 8.0%	0.8% / 1.6%	0.8% / 0.2%	0.0% / 0.0%	0.0% / 0.0%
0.5	1.1% / 8.0%	0.8% / 1.6%	0.8% / 0.2%	0.0% / 0.0%	0.0% / 0.0%

Exploratory conjunctions (not preregistered):

explanatory variable(s) (exploratory, post hoc)	median R^2	units with $R^2 > 0.3$	units with $R^2 > 0.5$
place (lattice node)	0.127	11.7%	2.5%
place x heading	0.202	30.5%	7.4%
place x frightened	0.165	23.4%	6.0%
pellets left (20 bins) x frightened	0.022	2.7%	1.1%
danger distance (20 bins) x danger direction	0.018	0.3%	0.0%

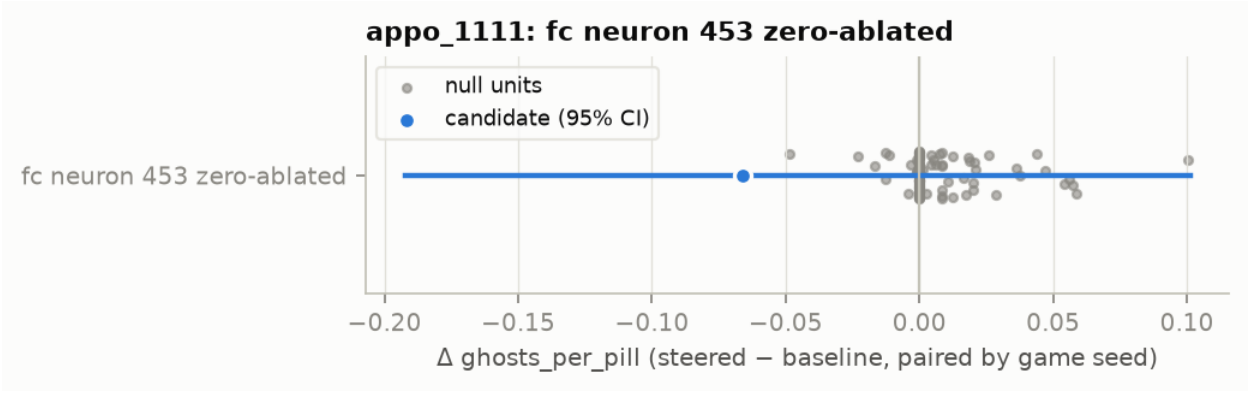
4.3 H2: steering

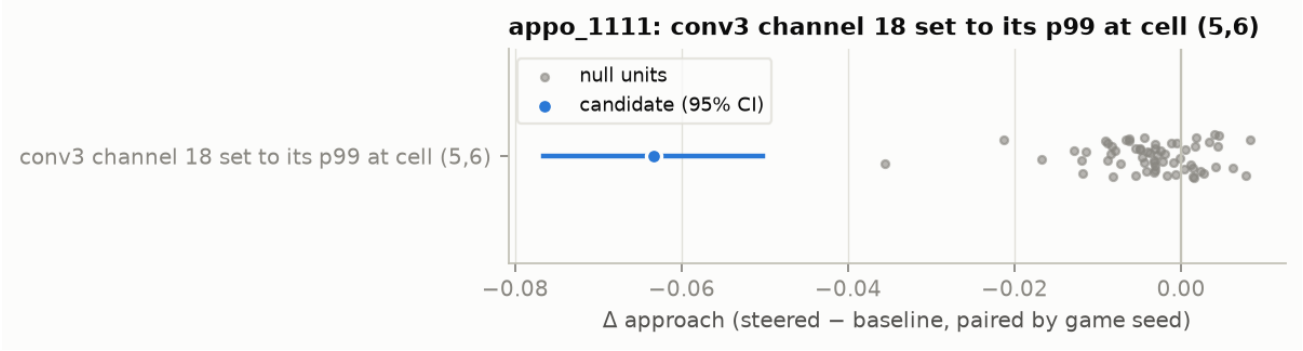
intervention	primary metric (predicted sign)	baseline → steered	Δ [95% CI]	Holm p	above null 95th pct?	H2
fc neuron 453 zero-ablated	ghosts_per_pill (−)	0.294 → 0.228	-0.066 [-0.192, 0.101]	0.2280	yes (null q95 0.013)	fail
conv3 channel 18 set to its p99 at cell (5,6)	approach (−)	0.424 → 0.361	-0.063 [-0.077, -0.050]	0.0002	yes (null q95 0.013)	pass
fc neuron 188 clamped to its p99 (constant logit bias, DLA UP -0.70) (positive control)	action_given_trigger (−)	0.395 → 0.305	-0.090 [-0.102, -0.079]	–	–	–

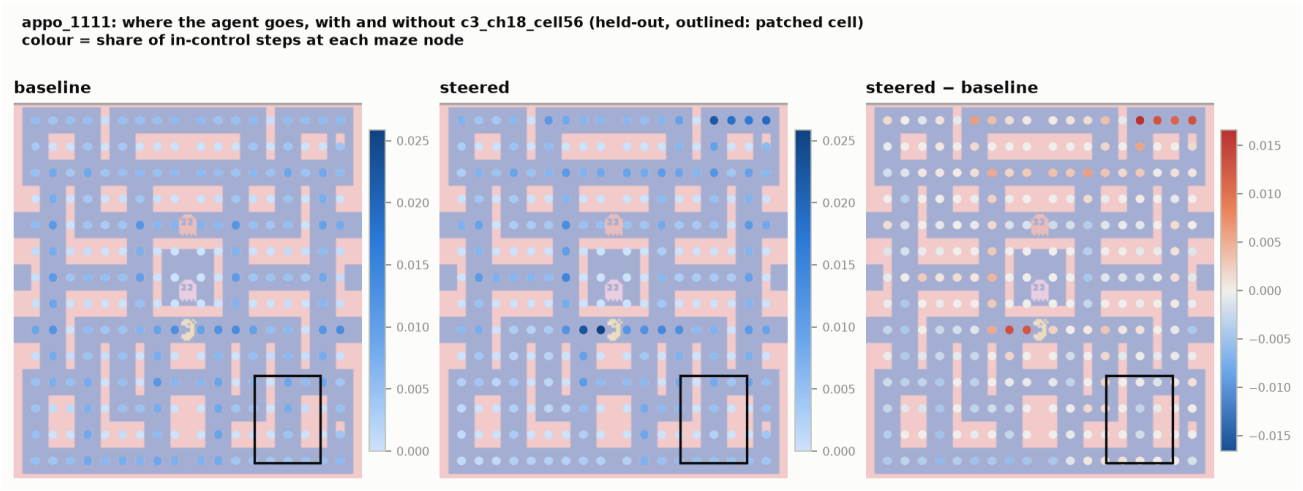
Off-target effects (two-sided paired sign-flip tests, Benjamini-Hochberg-corrected):

intervention	off-target metric	Δ [95% CI]	two-sided p	BH q
fc neuron 453 zero-ablated	return	-1619 [-2194, -1056]	0.0001	0.0001
fc neuron 453 zero-ablated	deaths_per_1k	1.14 [0.75, 1.52]	0.0001	0.0001
fc neuron 453 zero-ablated	pellets_per_1k	-9.23 [-17.68, -0.84]	0.0416	0.0468
fc neuron 453 zero-ablated	in_control	-469 [-633, -304]	0.0001	0.0001
conv3 channel 18 set to its p99 at cell (5,6)	return	-2490 [-2990, -2005]	0.0001	0.0001
conv3 channel 18 set to its p99 at cell (5,6)	deaths_per_1k	4.17 [3.27, 5.14]	0.0001	0.0001
conv3 channel 18 set to its p99 at cell (5,6)	pellets_per_1k	-12.16 [-20.78, -3.83]	0.0071	0.0091

intervention	off-target metric	Δ [95% CI]	two-sided p	BH q
conv3 channel 18 set to its p99 at cell (5,6)	ghosts_per_pill	-0.05 [-0.18, 0.09]	0.5156	0.5156
conv3 channel 18 set to its p99 at cell (5,6)	in_control	-788 [-944, -633]	0.0001	0.0001

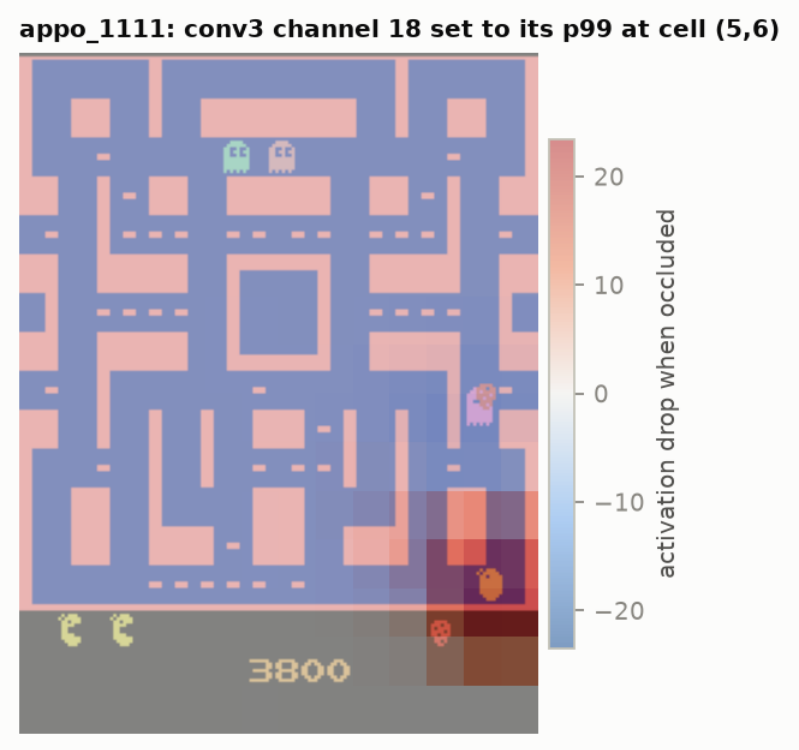


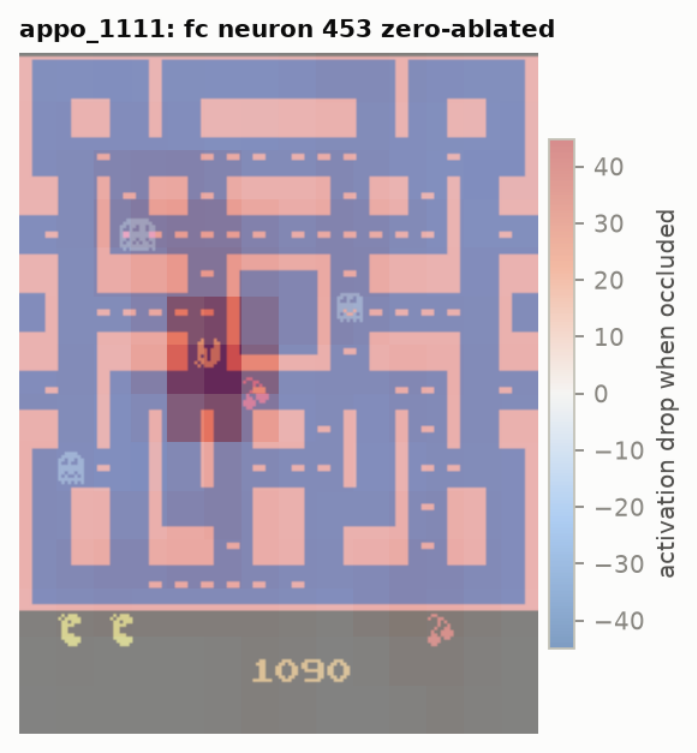




Attribution of the candidate units: exact upstream contributions, occlusion peak, and confound-controlled label checks (stratified AUROC within frightened × player-region strata; unique R^2 = drop-one loss of the gradient-boosting ceiling):

unit	strongest upstream units (mean exact contribution)	occlusion peak (screen x, y)	label checks: single-feature R^2 / stratified AUROC / unique R^2
fc neuron 453 zero-ablated	conv3 ch 17 (+5.1), conv3 ch 59 (+4.4), conv3 ch 1 (+4.3)	(57, 90)	dir_edible_up: 0.14 / 0.59 / 0.003; n_pellets: 0.38 / – / 0.007; n_edible: 0.29 / – / 0.002
conv3 channel 18 set to its p99 at cell (5,6)	conv2 ch 40 (+6.9), conv2 ch 12 (+5.5), conv2 ch 56 (+5.1)	(137, 165)	–





The no-go-zone unit depends on input inside its own receptive field. fc 453's "edible ghost is up" label does not survive confound control, so its ablation effect is attributed to its UP vote rather than to a ghost-direction representation.

Discovery-phase evidence that led to the candidates (exploratory):

rank	conv3 channel @ cell	Δ P(move toward cell) at junctions	dose (p99)
repel 1	ch 18 @ (5, 6)	-5.8 pp	24.6
repel 2	ch 15 @ (0, 6)	-5.1 pp	27.2
repel 3	ch 40 @ (4, 6)	-4.0 pp	34.8
repel 4	ch 36 @ (5, 6)	-3.6 pp	22.7
repel 5	ch 58 @ (5, 6)	-3.6 pp	18.4
attract 1	ch 16 @ (6, 2)	+3.6 pp	23.0
attract 2	ch 8 @ (1, 0)	+3.0 pp	21.6
attract 3	ch 38 @ (5, 0)	+3.0 pp	20.0
attract 4	ch 34 @ (6, 0)	+2.7 pp	28.1
attract 5	ch 46 @ (6, 0)	+2.6 pp	14.5

intervention (discovery pilot, paired games)	targeted metric	baseline → steered	Δ [95% CI]	Δ return	Δ deaths / 1k
`c3_ch20_ablate`	deaths_per_1k	2.536 → 2.730	0.194 [-0.183, 0.671]	-210	0.19
`c3_ch20_clamp_frightened`	ghosts_per_pill	0.392 → 0.213	-0.179 [-0.292, -0.060]	-3668	5.28
`c3_ch20_clamp`	deaths_per_1k	2.536 → 9.071	6.535 [5.265, 7.759]	-4047	6.54
`c3_ch39_ablate`	deaths_per_1k	2.536 → 4.169	1.633 [0.646, 2.502]	-2566	1.63

intervention (discovery pilot, paired games)	targeted metric	baseline → steered	Δ [95% CI]	Δ return	Δ deaths / 1k
`c3_ch53_ablate`	deaths_per_1k	2.536 → 2.665	0.129 [-0.035, 0.305]	-229	0.13
`c3_ch18_cell56`	approach	0.430 → 0.375	-0.056 [-0.068, -0.044]	-3520	5.57
`fc378_ablate`	action_given_trigger	0.502 → 0.475	-0.027 [-0.055, -0.000]	-1307	0.67
`fc453_ablate`	action_given_trigger	0.483 → 0.298	-0.184 [-0.204, -0.164]	-2739	1.52
`fc188_clamp`	action_given_trigger	0.384 → 0.302	-0.082 [-0.091, -0.072]	-3047	1.43

4.4 H3: replication

rule (frozen on agent 1111)	agent	selected unit	primary metric Δ [95% CI]	Holm p (within agent)	null units at least as strong	passes
zero-ablate the fc unit most needed for approaching edible ghosts	appo_2222	fc neuron 379 zero-ablated	-0.101 [-0.167, -0.036]	0.0025	0 of 99	yes
zero-ablate the fc unit most needed for approaching edible ghosts	appo_3333	fc neuron 351 zero-ablated	0.190 [0.071, 0.301]	0.9992	99 of 99	no
set the most repelling conv3 (channel, cell) to its p99	appo_2222	conv3 channel 3 set to its p99 at cell (0, 0)	-0.034 [-0.043, -0.026]	0.0002	0 of 63	yes
set the most repelling conv3 (channel, cell) to its p99	appo_3333	conv3 channel 46 set to its p99 at cell (5, 6)	-0.216 [-0.232, -0.200]	0.0002	0 of 63	yes

Steering per agent (held-out):

intervention	primary metric (predicted sign)	baseline → steered	Δ [95% CI]	Holm p	above null 95th pct?	H2
fc neuron 379 zero-ablated	ghosts_per_pill (−)	0.463 → 0.362	-0.101 [-0.167, -0.036]	0.0025	yes (null q95 0.027)	pass

intervention	primary metric (predicted sign)	baseline → steered	Δ [95% CI]	Holm p	above null 95th pct?	H2
conv3 channel 3 set to its p99 at cell (0, 0)	approach (−)	0.430 → 0.396	-0.034 [-0.043, -0.026]	0.0002	yes (null q95 0.007)	pass

intervention	primary metric (predicted sign)	baseline → steered	Δ [95% CI]	Holm p	above null 95th pct?	H2
fc neuron 351 zero-ablated	ghosts_per_pill (−)	0.660 → 0.850	0.190 [0.071, 0.301]	0.9992	no (null q95 0.019)	fail
conv3 channel 46 set to its p99 at cell (5, 6)	approach (−)	0.440 → 0.224	-0.216 [-0.232, -0.200]	0.0002	yes (null q95 0.012)	pass

5. Discussion

Are the neurons clean?

- **Not under single-feature labels.** With a 40-feature dictionary of single game variables, almost no fc neuron of these agents has one main job, and the agents are no cleaner than random directions in a meaningful way.
- **The basis is still not arbitrary.**
  - Random directions score *higher* best-single-feature R^2 than neurons (Fig. R^2). Dense mixtures pick up slow global variables such as pellets remaining.
  - Neurons are sparse (they fire on a minority of steps) and conjunctive: position × heading explains them far better than any single fact.
- **What the units are.** They resemble place cells more than "ghost detectors". This is consistent with saliency studies of Ms. Pac-Man agents, which found attention on corridors rather than ghosts (Greydanus et al. 2018).
- **Reconciling with Goodfire.** Goodfire's labels are free-text visual descriptions, which can express such conjunctions. Our result is a statement about a ground-truth *variable* basis. The conjunction analysis is post hoc and should be tested with a preregistered conjunctive dictionary.

Can single units steer behaviour?

- **Yes, causally and specifically, but not cleanly.** A single conv3 unit's activity is enough to repel the agent from a maze region. No other channel at that position does this at matched dose, and the same selection rule finds such a unit in an independently trained agent (H3).
- **Steering is destructive.** The agent abandons that region's pellets and dies more. Single-unit steering here is a blunt instrument, not a precise control knob.
- **Mechanism vs outcome.** The fc ablation shows a gap between mechanism and outcome. The predicted local behaviour ("go toward the edible ghost in direction d") changes strongly in agents 1111 and 3333. The global outcome (ghosts eaten per pill) changes little in agent 1111, falls in agent 2222, and *rises* in agent 3333. Removing one direction's vote reroutes the agent rather than simply weakening ghost hunting.
- **Why the fc layer is constrained.** The linear, recurrence-free policy head means fc-unit steering can only add fixed action biases. State-dependent steering must enter upstream, where activations pass through the fc ReLUs.

Methodological lessons.

1. Bit-exact fidelity tests are cheap and necessary. Library defaults changed under us (gymnasium's grayscale).

2. Near-deterministic policies need injected stochasticity (sticky actions) before games can serve as independent samples.
3. Discovery-pilot effects shrink on held-out data (winner's curse). Confirmatory effect sizes must come from held-out games.
4. A preregistered difference test needs a minimum effect size. Our H1 criterion was "passed" by a 0.6 percentage-point difference.

6. Limitations

- **Planned but not done.**
  - An input-pixel counterfactual for the top candidates (repainting ghosts as frightened in the observation) to separate ghost appearance from correlated game state.
  - A per-ghost "visible in the observation" flag (Atari sprite flicker can hide a ghost that RAM reports).
  - Training dynamics over the 36 milestone checkpoints of agent 1111.
  - Replication on agents trained with other algorithms (Stable-Baselines3 DQN/PPO).
  - The OCAtari cross-check of RAM decoding, replaced by pixel-level validation because OCAtari's PyPI package does not build.
- **Feature dictionary.** The one-job criterion depends on a finite 40-feature dictionary of *single* variables, so conjunctive codes are counted as "not clean" by construction. The conjunction analysis is post hoc.
- **Sticky actions.** These change the agent's task (it was trained without them) and lower its score. All results describe the agent under p = 0.25.
- **Dependence between games.** Games that share a start state overlap for their first few hundred steps. We report a start-state-cluster bootstrap as a robustness check.
- **Scope of interventions.** Steering was tested at fixed doses (discovery p99) and only for the frozen intervention types; dose-response is exploratory.
- **Agents.** Three public agents of one architecture and algorithm. Generalisation to other algorithms (DQN, PPO with IMPALA-ResNet) is untested.

7. Conclusion

Clean? In three reward-trained Ms. Pac-Man agents, individual neurons are not "already clean" with respect to exact single game variables. They are sparse, conjunctive, place-like codes.

Steerable? Yes, causally and specifically. Single units steer behaviour beyond matched random units: a single conv3 unit creates a reproducible no-go zone, and a single fc unit gates a predicted piece of ghost-chasing. These effects come with substantial collateral damage.

Toolkit. The fidelity-checked models, RAM ground truth, preregistration and microscope make these agents a convenient testbed for future work: conjunctive dictionaries, sparse autoencoders, and other algorithms.

Appendix A. Fidelity details

See `reports/01_fidelity.md`.

Appendix B. Feature list

See `src/pacneuron/features.py` and `reports/02_ram_ground_truth.md`.

Appendix C. Preregistration and deviations

See `reports/preregistration.md`, `reports/preregistration_addendum.md` and `reports/00_summary.md`.

AI statement

This codebase and paper were written by Claude Code (Anthropic's Claude), following an exact specification from a human author who directed the research. Human cleanup and rewriting are in progress.